

Flow Diagram

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flowchart LR
    %% =====
    %% CLIENT
    %% =====
    subgraph CLIENT_ZONE["Client Layer"]
        CUSTOMER["Customer / Client"]
        ADMIN["Admin UI"]
    end

    %% =====
    %% GO FIBER SINGLE INSTANCE
    %% =====
    subgraph APP["Go Fiber Application - Single Instance"]

        subgraph API_LAYER["API and Middleware Layer"]
            ROUTER["Go Fiber Router"]

            PRICING_API["Pricing API<br/>POST /api/v1/pricing/calculate"]

            PROMOTION_API["Promotion Management API<br/>Create / Update / Activate Promotion"]

            MIDDLEWARE["Middleware<br/>Authentication<br/>Authorization<br/>Request Validation<br/>Request ID<br/>Error Handling"]
        end

        subgraph APPLICATION_LAYER["Application Service Layer"]
            PRICING_SERVICE["Pricing Service<br/>ควบคุม Calculation Flow"]

            PROMOTION_SERVICE["Promotion Service<br/>จัดการ Promotion Configuration"]

            LOG_SERVICE["Calculation Log Service<br/>สร้าง Audit Snapshot"]
        end

        subgraph REPOSITORY_LAYER["Repository Layer - GORM"]
            PRODUCT_REPO["Product Repository"]
            PROMOTION_REPO["Promotion Repository"]
            USAGE_REPO["Promotion Usage Repository"]
            ORDER_REPO["Order Repository"]
            LOG_REPO["Calculation Log Repository"]
        end

        subgraph ENGINE["Promotion Engine - Rule-based Engine"]

            CONTEXT["Calculation Context<br/>Products<br/>Quantities<br/>Current Amount<br/>"]
        end
    end
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r/>Coupon<br/>User Context"]

    ELIGIBILITY["Eligibility Evaluator<br/>Specification Pattern<br/>ตรวจ Target
และ Condition"]

    SORTER["Promotion Sorter<br/>เรียง Scope และ Priority"]

    STACKING["Stacking Policy<br/>Policy Pattern<br/>ตรวจ Stackable / Exclusive<br/>Conflict / Stop Processing"]

    REGISTRY["Strategy Registry<br/>Factory and Registry Pattern<br/>เลือก Strategy จาก Action Type"]

    subgraph STRATEGIES["Promotion Strategies"]
        PERCENTAGE["Percentage Discount Strategy<br/>ลดเป็นเปอร์เซ็นต์"]

        FIXED["Fixed Amount Strategy<br/>ลดเป็นจำนวนเงิน"]

        CART["Cart Discount Strategy<br/>ลดทั้ง Order"]

        SHIPPING["Free Shipping Strategy"]

        FUTURE["Future Promotion Strategy<br/>รองรับ Promotion รูปแบบใหม่"]
    end

    CAP["Discount Cap and Validation<br/>ส่วนลดไม่เกินยอดคงเหลือ<br/>Final Total ต้องไม่ติดลบ"]

    RESULT["Calculation Result Builder<br/>Original Total<br/>Discount Total<br/>Final Total<br/>Applied Promotions<br/>Skipped Promotions<br/>Calculation Breakdown"]
    end
end

%% =====
%% MYSQL SINGLE INSTANCE
%% =====
subgraph MYSQL["MySQL Database - Single Instance"]

    subgraph PRODUCT_DATA["Product Data"]
        CATEGORIES[("product_categories")]
        PRODUCTS[("products")]
    end

    subgraph PROMOTION_CONFIG["Flexible Promotion Configuration"]
        PROMOTIONS[("promotions<br/>Scope / Priority<br/>Stacking Policy")]
        TARGETS[("promotion_targets<br/>Promotion ใช้กับอะไร")]
        CONDITIONS[("promotion_conditions<br/>ต้องผ่านเงื่อนไขอะไร")]
        ACTIONS[("promotion_actions<br/>ลดราคาอย่างไร")]
    end

    subgraph TRANSACTION_DATA["Transaction Data"]
        ORDERS[("orders")]
    end

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ORDER_ITEMS[("order_items")]
USAGES[("promotion_usages")]
end

subgraph AUDIT_DATA["Audit Data"]
    CALCULATION_LOGS[("promotion_calculation_logs<br/>Request Snapshot<br/>Applied / Skipped<br/>Calculation Breakdown")]
end
end

%% =====
%% CLIENT REQUEST FLOW
%% =====
CUSTOMER -->|"1. ส่งรายการสินค้า<br/>POST /api/v1/pricing/calculate"| ROUTER

ADMIN -->|"จัดการ Promotion"| ROUTER

ROUTER --> PRICING_API
ROUTER --> PROMOTION_API

PRICING_API -->|"2. ตรวจสอบ Request"| MIDDLEWARE
MIDDLEWARE -->|"3. Request ถูกต้อง"| PRICING_SERVICE

PROMOTION_API -->|"Create / Update / Activate"| PROMOTION_SERVICE
PROMOTION_SERVICE --> PROMOTION_REPO

%% =====
%% LOAD PRODUCT DATA
%% =====
PRICING_SERVICE -->|"4. โหลดราคาสินค้าจริง"| PRODUCT_REPO
PRODUCT_REPO -->|"Query Product"| PRODUCTS
PRODUCTS -->|"Product Price and Status"| PRODUCT_REPO
PRODUCT_REPO -->|"5. Product Data"| PRICING_SERVICE

CATEGORIES -->|"Category Relationship"| PRODUCTS

%% =====
%% LOAD PROMOTION DATA
%% =====
PRICING_SERVICE -->|"6. โหลด Active Promotions"| PROMOTION_REPO

PROMOTION_REPO --> PROMOTIONS
PROMOTIONS --> TARGETS
PROMOTIONS --> CONDITIONS
PROMOTIONS --> ACTIONS

TARGETS -->|"Targets"| PROMOTION_REPO
CONDITIONS -->|"Conditions"| PROMOTION_REPO
ACTIONS -->|"Actions"| PROMOTION_REPO
PROMOTIONS -->|"Promotion Policy"| PROMOTION_REPO

PROMOTION_REPO -->|"7. Promotion Configuration"| PRICING_SERVICE

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%% =====
%% CHECK USAGE LIMIT
%% =====
PRICING_SERVICE -->|"8. ตรวจสอบ Usage Limit"| USAGE_REPO
USAGE_REPO --> USAGES
USAGES -->|"Usage Information"| USAGE_REPO
USAGE_REPO -->|"9. Usage Result"| PRICING_SERVICE

%% =====
%% PROMOTION CALCULATION
%% =====
PRICING_SERVICE -->|"10. Build Calculation Context"| CONTEXT

CONTEXT -->|"11. ตรวจสอบ Promotion ที่ใช้ได้"| ELIGIBILITY

ELIGIBILITY -->|"ตรวจสอบ Product / Category / Cart Target"| TARGETS
ELIGIBILITY -->|"ตรวจสอบ Conditions"| CONDITIONS
ELIGIBILITY -->|"Eligible Promotions"| SORTER

SORTER -->|"12. เรียงลำดับ<br/>ITEM → CART → COUPON → SHIPPING<br/>Priority ASC"| STACKING

STACKING -->|"13. Promotion ที่ใช้ร่วมกันได้"| REGISTRY
STACKING -->|"Promotion ที่ Conflict"| RESULT

REGISTRY -->|"PERCENTAGE_DISCOUNT"| PERCENTAGE
REGISTRY -->|"FIXED_AMOUNT_DISCOUNT"| FIXED
REGISTRY -->|"CART_DISCOUNT"| CART
REGISTRY -->|"FREE_SHIPPING"| SHIPPING
REGISTRY -->|"Action Type ใหม่"| FUTURE

PERCENTAGE --> CAP
FIXED --> CAP
CART --> CAP
SHIPPING --> CAP
FUTURE --> CAP

CAP -->|"14. Validated Discount Result"| RESULT

%% =====
%% CALCULATION LOG FLOW
%% =====
RESULT -->|"15. Final Price and Breakdown"| PRICING_SERVICE

PRICING_SERVICE -->|"16. สร้าง Calculation Snapshot"| LOG_SERVICE

LOG_SERVICE -->|"บันทึก Log ผ่าน Repository"| LOG_REPO
LOG_REPO -->|"Insert Calculation Log"| CALCULATION_LOGS
CALCULATION_LOGS -->|"Calculation ID"| LOG_REPO
LOG_REPO -->|"18. Log Saved"| LOG_SERVICE
LOG_SERVICE -->|"Calculation ID"| PRICING_SERVICE

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%% =====
%% RESPONSE FLOW
%% =====
PRICING_SERVICE -->|"19. Calculation Result"| PRICING_API

PRICING_API -->|"20. ส่งผลลัพธ์<br/>Final Price<br/>Applied Promotions<br/>Skipped Promotions<br/>Calculation ID"| CUSTOMER

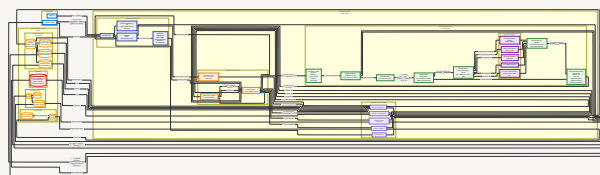
%% =====
%% ORDER CONFIRM FLOW
%% =====
CUSTOMER -->|"ยืนยัน Order"| ROUTER
ROUTER --> ORDER_REPO

ORDER_REPO --> ORDERS
ORDERS --> ORDER_ITEMS
ORDERS --> USAGES

%% =====
%% STYLING
%% =====
classDef client fill:#E3F2FD,stroke:#1976D2,color:#000;
classDef api fill:#E8EAF6,stroke:#3F51B5,color:#000;
classDef service fill:#FFF3E0,stroke:#F57C00,color:#000;
classDef engine fill:#E8F5E9,stroke:#388E3C,color:#000;
classDef strategy fill:#F3E5F5,stroke:#7B1FA2,color:#000;
classDef database fill:#FFFDE7,stroke:#F9A825,color:#000;
classDef audit fill:#FFE0B2,stroke:#D32F2F,color:#000;

class CUSTOMER,ADMIN client;
class ROUTER,PRICING_API,PROMOTION_API,MIDDLEWARE api;
class PRICING_SERVICE,PROMOTION_SERVICE,LOG_SERVICE service;
class CONTEXT,ELIGIBILITY,SORTER,STACKING,REGISTRY,CAP,RESULT engine;
class PERCENTAGE,FIXED,CART,SHIPPING,FUTURE strategy;
class CATEGORIES,PRODUCTS,PROMOTIONS,TARGETS,CONDITIONS,ACTIONS,ORDERS,ORDER_ITEMS,USAGES database;
class CALCULATION_LOGS audit;

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flowchart LR

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A["Customer / Client<br/>ส่งรายการสินค้า"] --> B["Go Fiber API<br/>POST /api/v1/pricing/calculate"]

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B --> C["Middleware<br/>Validate Request<br/>Auth / Request ID / Error Handling"]

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C --> D["Pricing Service<br/>ควบคุม Flow การคำนวณราคา"]

D --> E["Product Repository<br/>โหลดราคาสินค้าจริง"]
E --> F[("MySQL<br/>products<br/>product_categories")]
F --> E
E --> D

D --> G["Promotion Repository<br/>โหลด Promotion ที่ Active"]
G --> H[("MySQL<br/>promotions<br/>promotion_targets<br/>promotion_conditions<br/>promotion_actions")]
H --> G
G --> D

D --> I["Promotion Engine<br/>คำนวณ Promotion"]

I --> J["1. Check Target<br/>สินค้า / หมวดหมู่ / Cart"]
J --> K["2. Check Condition<br/>ยอดขั้นต่ำ / Coupon / วันที่ใช้งาน"]
K --> L["3. Sort Promotion<br/>ITEM → CART → COUPON → SHIPPING<br/>priority น้อยทำก่อน"]
L --> M["4. Stacking Policy<br/>stackable / exclusive / stop_processing"]
M --> N["5. Strategy Pattern<br/>เลือกวิธีลดราคา"]

N --> O["Percentage Strategy<br/>ลดเป็น %"]
N --> P["Fixed Amount Strategy<br/>ลดเป็นจำนวนเงิน"]
N --> Q["Future Strategy<br/>Promotion ใหม่ในอนาคต"]

O --> R["Calculate Result"]
P --> R
Q --> R

R --> S["Validate Discount<br/>ห้ามลดเกินราคา<br/>final_total ต้องไม่ติดลบ"]

S --> T["Calculation Result<br/>original_total<br/>discount_total<br/>final_total<br/>applied_promotions<br/>skipped_promotions"]

T --> U["Calculation Log Service<br/>สร้าง Snapshot การคำนวณ"]

U --> V[("MySQL<br/>promotion_calculation_logs")]
V --> U

U --> D
D --> W["Go Fiber API<br/>Response"]
W --> X["Customer / Client<br/>ได้รับ Final Price + Breakdown"]

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flowchart LR

A["Customer / Client"] --> B["Go Fiber API"]

B --> C["Pricing Service"]

C --> D["Promotion Engine"]

D --> E["Calculation Result"]

E --> F["Calculation Log Service"]

C --> DB

D --> DB

F --> DB

subgraph DB["MySQL Database - Single Instance"]

T1["Product Tables
products
product_categories"]

T2["Promotion Tables
promotions
promotion_targets
promotion_conditions
promotion_actions"]

T3["Order Tables
orders
order_items"]

T4["Usage Table
promotion_usages"]

T5["Audit Table
promotion_calculation_logs"]

end

F --> G["Response to Customer"]

